Predicting Diabetes and BMI with Deep Learning

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Background

- The project builds machine learning models to predict diabetes status and BMI from lifestyle and health indicators using the CDC BRFSS 2015 dataset.

Research Questions

- **Classification:** Can we predict if patient has a diabetes or not based on lifestyle factors like BMI, Blood Pressure, cholesterol levels… and physical activity ?
- **Regression:** Can we predict a patient's BMI based on lifestyle factors like age, income, physical activity, general health rating, and smoking habits?

Dataset

- **Single Dataset :** CDC Diabetes Health Indicators (UCI ML Repository) , 253,680 instances, 22 features, No missing values, Binary, categorical, and ordinal features

Data Preparation

- Classification: Applied random undersampling to balance classes (35,346 diabetic vs 35,346 non-diabetic). StandardScaler applied. 80/20 stratified split.
- Regression: Used full dataset (253,680 rows). StandardScaler applied. 80/20 split.

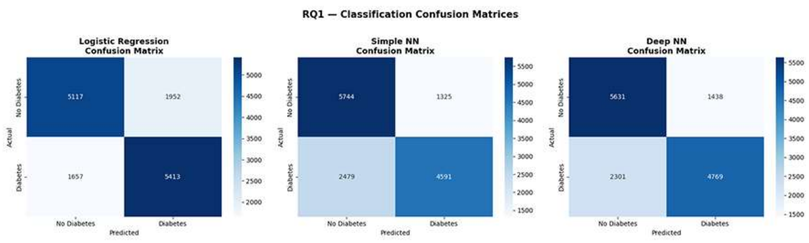
Classification Method

- Logistic Regression was established as the baseline model.
- Deep learning models were developed using a Simple Feedforward Neural Network (64→32→1) and a Deep FNN incorporating Batch Normalization and Dropout layers.
- Tuned Learning Rate, Dropout Rate, and Batch Size to explicitly maximize Recall, minimizing missed diabetes diagnoses.

Classification Results

- Logistic Regression achieved the highest Recall of **76.6%**, correctly identifying 3 out of 4 diabetic patients in the test set.
- Precision of **73.5%** is an acceptable clinical trade-off, as maximizing diabetes detection was the primary goal.

Model	Recall	F1 Score
Logistic Regression	0.766	0.750
Simple NN	0.638	0.702
Deep NN	0.687	0.724



Regression Method

- Linear Regression was established as the baseline model.
- Deep learning models were developed using a Simple Feedforward NN (64→32→1) and a Deep FNN (256→128→64→1) incorporating Batch Normalization and Dropout layers.
- Tuned Learning Rate, Dropout Rate, and Batch Size to minimize MAE and improve BMI prediction accuracy.

Regression Results

- Deep NN achieved the best MAE of **4.2255** while Simple NN achieved the best R^2 of **0.1599**, both outperforming the Linear Regression baseline.
- All models clustered predictions around the population mean BMI (~28-30), reflecting a data ceiling caused by the binary nature of input features.

Model	MAE	R^2
LR	4.4012	0.1193
SNN	4.2618	0.1599
DNN	4.2255	0.1577

Future Work

- Future studies will include using SMOTE oversampling for better diabetes recall as well as acquiring better continuous attributes such as caloric consumption and physical exercise hours to surpass the BMI prediction limitation of R^2 ~0.16.